

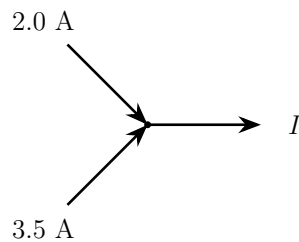
Written HW 5 — Circuits I

Ohm's law, the junction rule, the loop rule, and series & parallel resistors. **Section 1 is required: 9 problems, 2 points each (18 points total).** **Section 2 is extra credit: 9 problems, 1 point each (up to 9 extra points).** Every battery below is ideal ($r = 0$). Show your work.

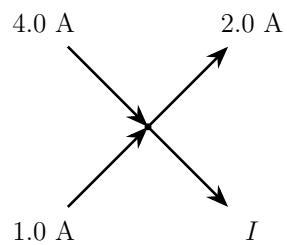
Part 1 — Required — 2 points each

1. A resistor has resistance $R = 25\ \Omega$. A voltmeter placed across it reads $\Delta V = 15\ \text{V}$. What current flows through the resistor?

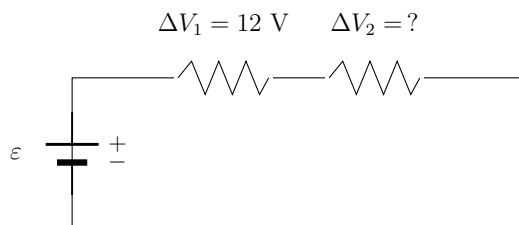
2. Three wires meet at the junction shown below. Find the current I leaving the junction.



3. Four wires meet at the junction shown below. Find the unknown current I .

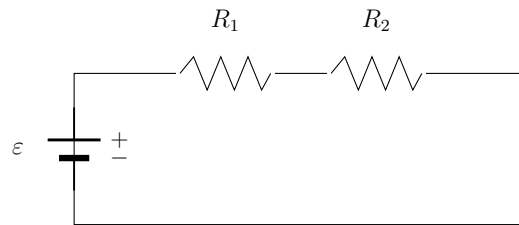


4. The single-loop circuit below has a battery with $\varepsilon = 20$ V and two resistors in series. A voltmeter reads $\Delta V_1 = 12$ V across the first resistor. Find ΔV_2 across the second.



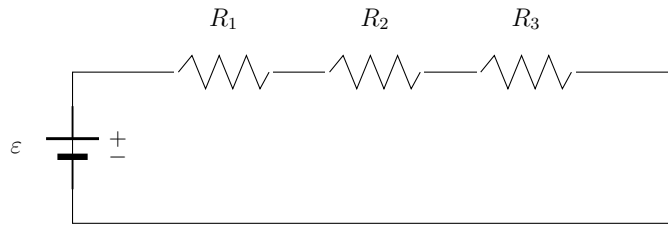
5. Same idea, different numbers: a single loop has $\varepsilon = 24$ V and two series resistors. $\Delta V_1 = 9.0$ V. Find ΔV_2 .

6. Two resistors, $R_1 = 4.0\ \Omega$ and $R_2 = 8.0\ \Omega$, are in series with a battery of $\varepsilon = 18$ V.



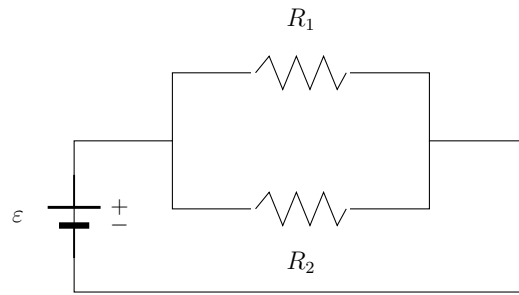
Find (a) the equivalent resistance, and (b) the current supplied by the battery.

7. Three resistors, $R_1 = 2.0\ \Omega$, $R_2 = 3.0\ \Omega$, and $R_3 = 5.0\ \Omega$, are in series with a battery of $\varepsilon = 20\ \text{V}$.



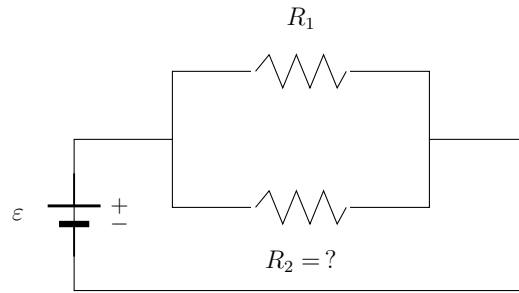
Find (a) the equivalent resistance, and (b) the current supplied by the battery.

8. Two resistors, $R_1 = 6.0\ \Omega$ and $R_2 = 3.0\ \Omega$, are in parallel across a battery of $\varepsilon = 12\ \text{V}$.



Find (a) the equivalent resistance, and (b) the total current from the battery.

9. Two resistors are in parallel across a battery of $\varepsilon = 12\text{ V}$. $R_1 = 6.0\ \Omega$, and the *total* current measured from the battery is 5.0 A .

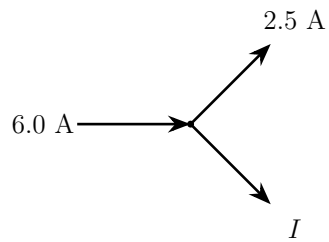


Find (a) the current through R_2 , and (b) the value of R_2 .

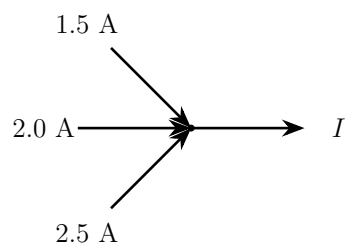
Part 2 — Extra credit — 1 point each

10. A current of 0.40 A flows through a resistor when a voltmeter across it reads 8.0 V . Find R .

- 11.** At the junction shown below, 6.0 A flows in and splits into two outgoing wires; one carries 2.5 A. Find the other.



- 12.** Three wires carry current *into* the junction shown below: 1.5 A, 2.0 A, and 2.5 A. One wire carries it all back out. Find that current.



13. A single loop has $\varepsilon = 9.0$ V and *three* series resistors. Voltmeters across two of them read 2.0 V and 3.0 V. Find the reading across the third.

14. A single loop has $\varepsilon = 30$ V and three series resistors. $\Delta V_1 = 10$ V and $\Delta V_2 = 8.0$ V. Find ΔV_3 .

15. Two resistors, $R_1 = 3.0\ \Omega$ and $R_2 = 7.0\ \Omega$, are in series with a battery of $\varepsilon = 20$ V. Find the current supplied by the battery.

16. Two resistors are in series with a battery of $\varepsilon = 15 \text{ V}$. $R_1 = 2.0 \, \Omega$, and the current measured in the circuit is 1.5 A . Find R_2 .

17. Two resistors, $R_1 = 8.0 \, \Omega$ and $R_2 = 24 \, \Omega$, are in parallel across a battery of $\varepsilon = 24 \text{ V}$. Find the total current supplied by the battery.

18. Two resistors, $R_1 = 5.0 \, \Omega$ and $R_2 = 20 \, \Omega$, are in parallel across a battery. The *total* current from the battery is measured to be 5.0 A . Find ε .